



PSOC™ 4 CAPSENSE™ using PSOC™ 4000T and PSOC™ 4100T Plus

Customer training workshop (CTW)

May 2026



Prerequisites

- Basic hands-on with ModusToolbox™
- Basic understanding of PSOC™ 4xxx device architecture
- Install or update the following on your PC:
 - ModusToolbox™ Tools Package v3.5.0
 - Python v3.12.x (Python v3.8.x to v3.12.x are supported by ModusToolbox™ v3.5.0)
 - ModusToolbox™ Eclipse IDE 3.5
 - ModusToolbox™ Programming Tools 1.4
 - ModusToolbox™ CAPSENSE™ and Multi-sense Pack

Agenda

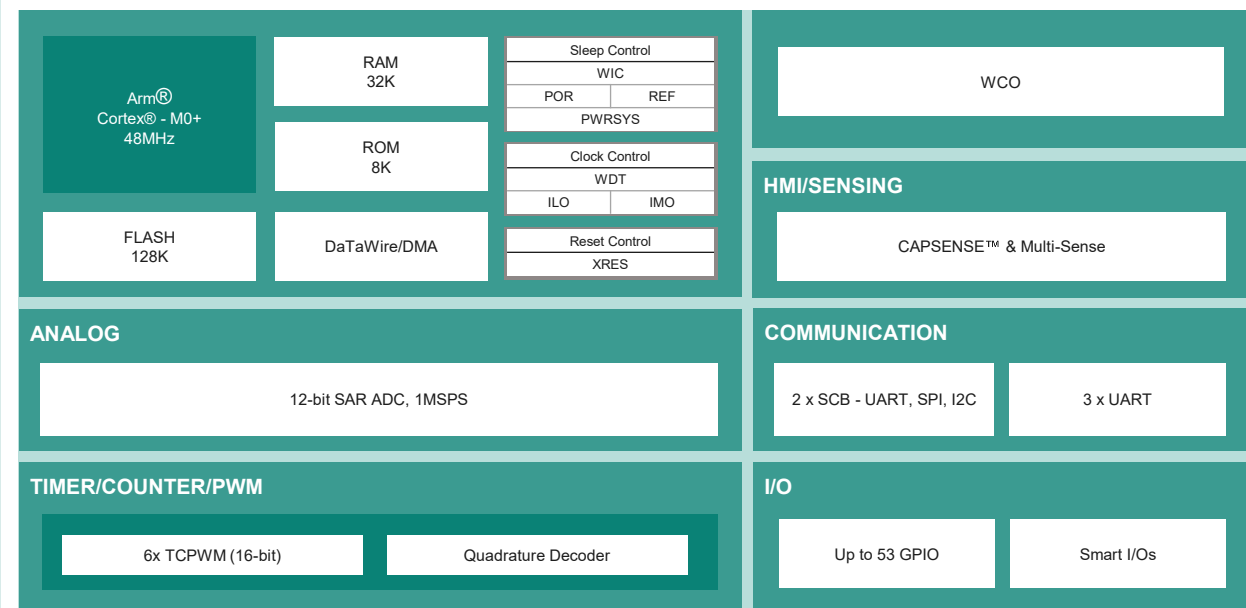
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2	What's new in MSCLP	8
3	Low-power CAPSENSE™ design	14
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32-bit MCU for multi-sense control

Block diagram – PSOC™ 4100T Plus



Applications

Home appliances, smart home, wearables, hearables, and other industrial and consumer applications.

Features

32-bit MCU subsystem

- 48 MHz Arm® Cortex®-M0+ CPU
- 128 KB flash and 32 KB SRAM
- DMA

Multi-sense with next-generation CAPSENSE™

- Fifth Gen CAPSENSE™ block with multi-sense, "Always-ON" sensing enabling ultra-low-power consumption
- Supports self-cap, mutual-cap and inductive sensing technologies
- Supports up to 32 sensors

Programmable digital blocks

- Six 16-bit timer/counter/pulse-width modulator (TCPWM) blocks
- Two serial communication blocks (SCBs) that are configurable as I²C, SPI, or UART
- Three dedicated UART interfaces

I/O subsystem

- Up to 53 GPIOs, including 32 sensors

Packages

- 48-QFN, 44-TQFP, 48-TQFP, and 64-TQFP

Other

- Main clock (IMO) 48 MHz with $\pm 1\%$ accuracy
- Enhanced ESD: 6.5 KV HBM, 2 KV CDM

CAPSENSE™ success with MSCLP 5th Gen

Parameter	CSDv2 (4 th Gen)	MSCv3 (5th Gen) PSOC™ 4100S Max	MSCLP (5 th Gen with low-power)		Improvements
			PSOC™4000T	PSOC™4100T Plus	
Converter resolution (ENOB – Effective number of bits)	10-bit	14-bit	14-bit	14-bit	MSCv3, MSCLP 10x high SNR enables: – High performance sensors – Long distance proximity and SAR sensor – Interoperable and scalable low-cost touchscreen – High performance small screens mid-range performance large screens – Complex UI interfaces
External noise immunity	Medium	High	High	High	–
Autonomous scan (DMA, RAM)	N/A	Yes (RAM / DMA)	No DMA	Yes (RAM / DMA*)	~40% more CPU bandwidth for application processing (PSOC™ 4) ~20% faster HMI touch response rate
Multi-channel sensing	N/A	Yes	N/A	N/A	Enable large and complex HMIs (on connected devices)
Multi-chip sensing	N/A	Yes	N/A	N/A	Enable large and complex HMIs (on industrial large screens)
Wake-on-touch (WoT) / LP sensing	N/A	N/A	Yes	Yes	10x improvement of power consumption Enable low-power applications in wearable segment
Additional features		CIC2, HW IIR, CDAC Dither	All of MSCV3, Fine CDAC	All of MSCV3, Fine CDAC	–
CIC2	NA	Acc- 20-sample – 12-bit	Acc- 24-bit sample – 16-bit	Acc- 24-bit sample – 16-bit	–
External frame start	N/A	Yes	N/A	Yes	External Frame Start is add-on feature which helps to sync CAPSENSE™ scan with an external trigger.

* - With Application Layer Handling

Capacitive sensing technology- Comparison

Parameters	CSD V1 (3 rd Gen)	CSD V2 (4 th Gen)	MSCLP (5 th Gen)	Competition
Maximum supported CSD capacitance to achieve 5:1 SNR – hardware only	< 31.5 pF	< 37.3 pF	<230 pF	28 pF for competition with best SNR
SNR (CSD)	5:1	6.5:1	48:1	–
Power supply rejection (DC)	-59 dB (GPIO)	-61 dB (GPIO)	-80 dB (GPIO)	N/A
Common mode rejection (AC)	N/A	-47 dB	-70 dB	N/A
Overlay thickness supported (CSD)	< 5 mm	5mm	18 mm	N/A
Active average current	–	156 μ A	117.8 μ A	–
ALR average current	–	40 μ A	29.2 μ A	–
Wake-on-touch (WoT) average current	–	21 μ A	3.8 μ A	–

Notes:

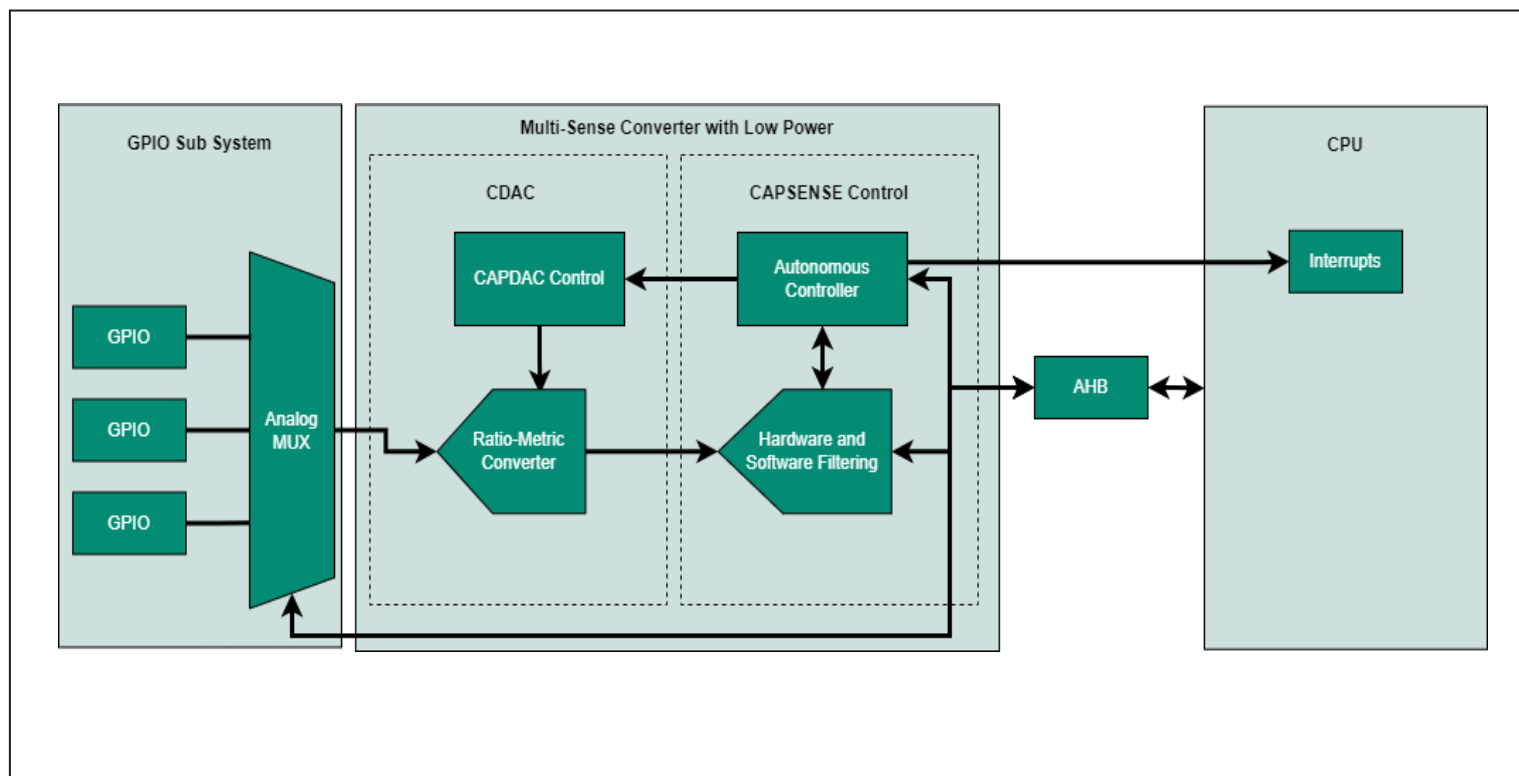
- Limited software filtering in the above silicon results. Optimal software filtering significantly improves performance (up to 3x for CSD noise floor based on 7 pF Si measurement).
- Key message: Optimal co-design of hardware and software, including tweaks for a custom application, based on several generations of experience.

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MSCLP 5th generation: Key updates

- Improved SNR
- Higher sensitivity
- Higher sensor capacitance range
- Easier layout
- Greater range of sensor sizes
- Wake-on-touch (WoT)
- Ultra-low-power autonomous operation
- Improved liquid tolerance
- External sync

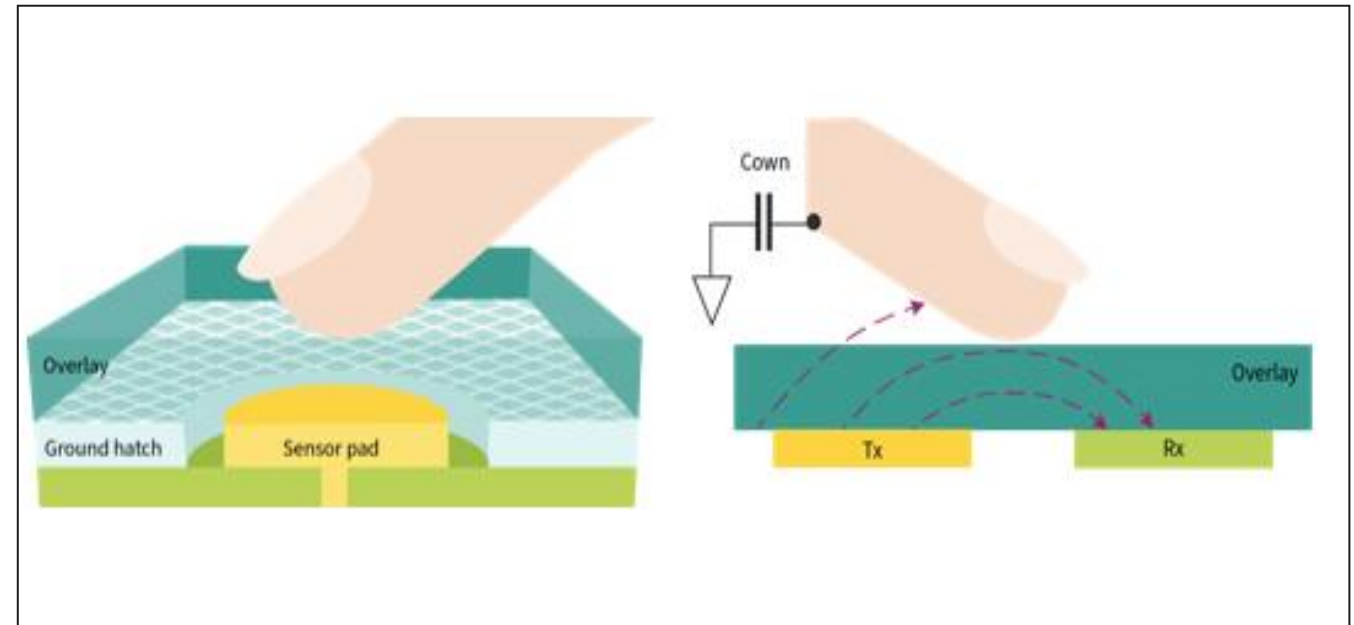


Improved SNR

- All new ratio-metric analog architecture
- Advanced hardware filtering
- Inherently lower internal noise sources
 - System noise floor: 0.1 fF-rms with 8 pF input capacitor
- **Benefits**
 - Shorter scan duration
 - Less software filtering
 - Lower power consumption
 - Robust operation in harsh environments
 - Ratio-metric design leading to improved DC PSRR

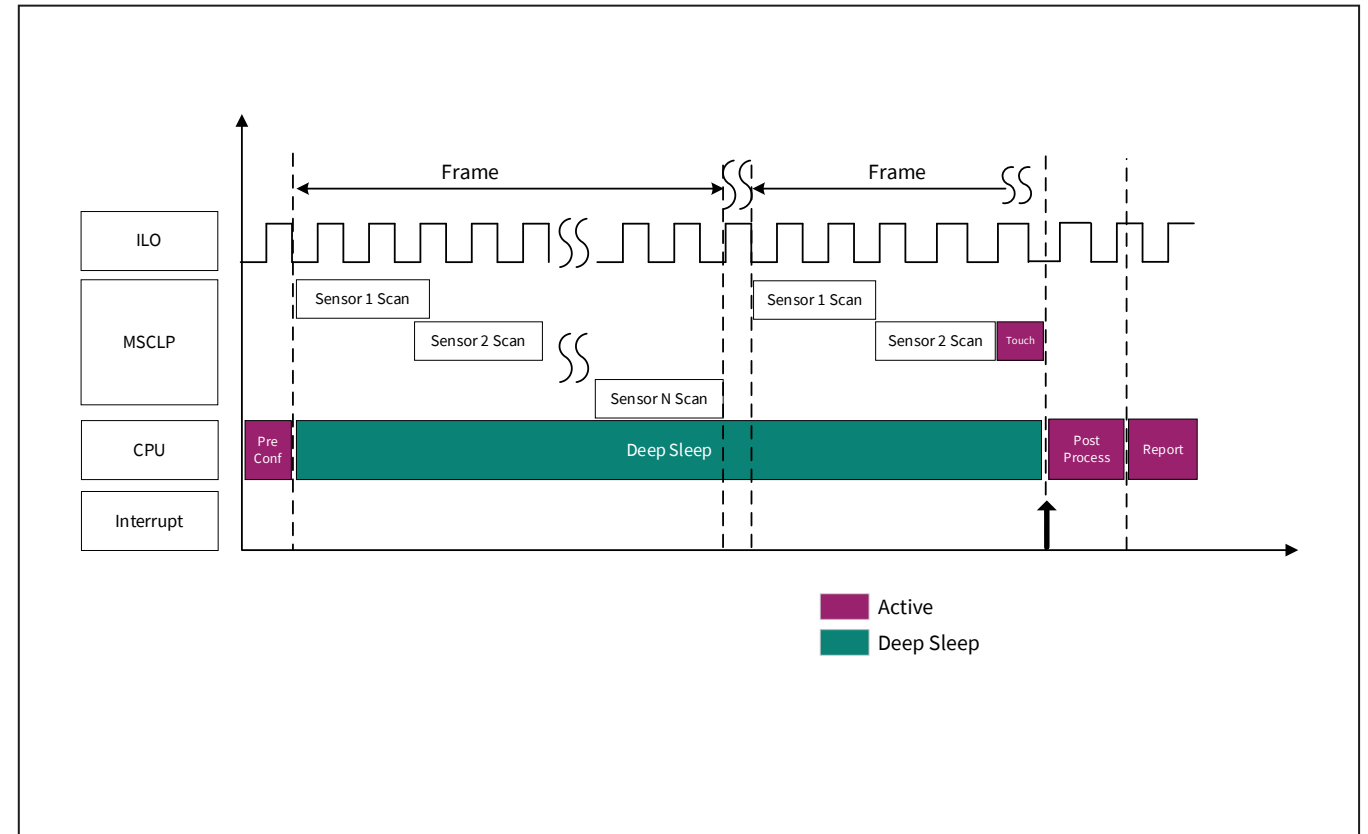
Higher sensitivity

- **Higher sensor capacitance range**
 - Input capacitance range for self-capacitance: 2 – 200 pF
 - Input capacitance range for mutual-capacitance: 0.5 – 30 pF
- **Benefits**
 - Support smaller and larger sensors
 - Higher proximity detection range
 - Much wider range of overlay thickness and materials



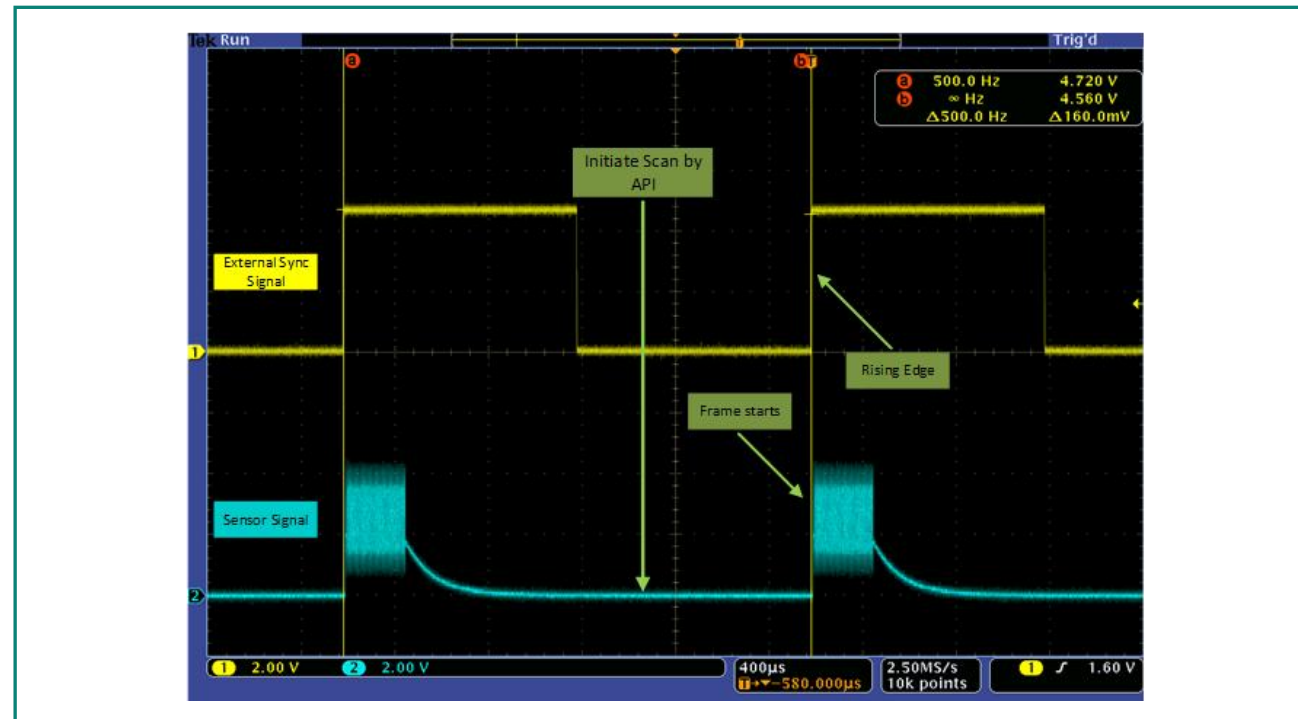
Wake-on-touch (WoT)

- Sensors scanned by HW at low refresh rate
- Results processed without CPU intervention
- Device can be kept in Deep Sleep (CPU, HFCLK OFF) while MSCLP is scanning
- Low-power interrupt generated by MSCLP hardware when touch is detected
- **Benefits**
 - Device can be in Deep Sleep when there is no touch activity
 - Lower-power consumption



External frame start

- Synchronize start of frame scan to an external signal
- Initiated by a rising edge on ext_frm_start pin
- External signal can even be non-periodic
- Benefits
 - Sync with external noise source (for example, LCD)
 - Avoid the noise or keep the effect consistent from scan to scan
 - Scanning rate controlled by host

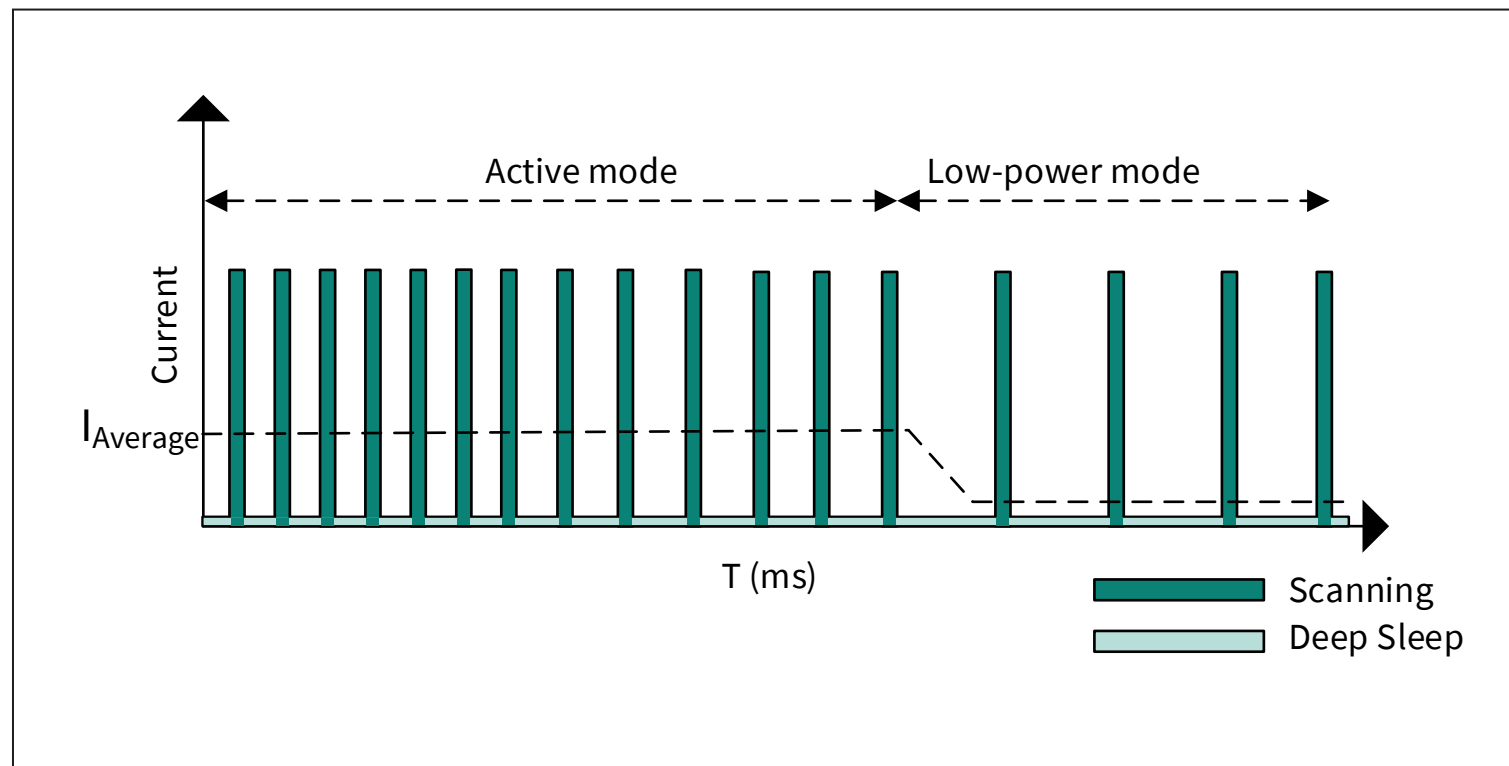


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Low-power CAPSENSE™ design

- Wearables are in low-power mode most of the time
- Current consumption in low-power mode significantly impacts battery life
- Touch detection is used to transition from the low-power to the active state

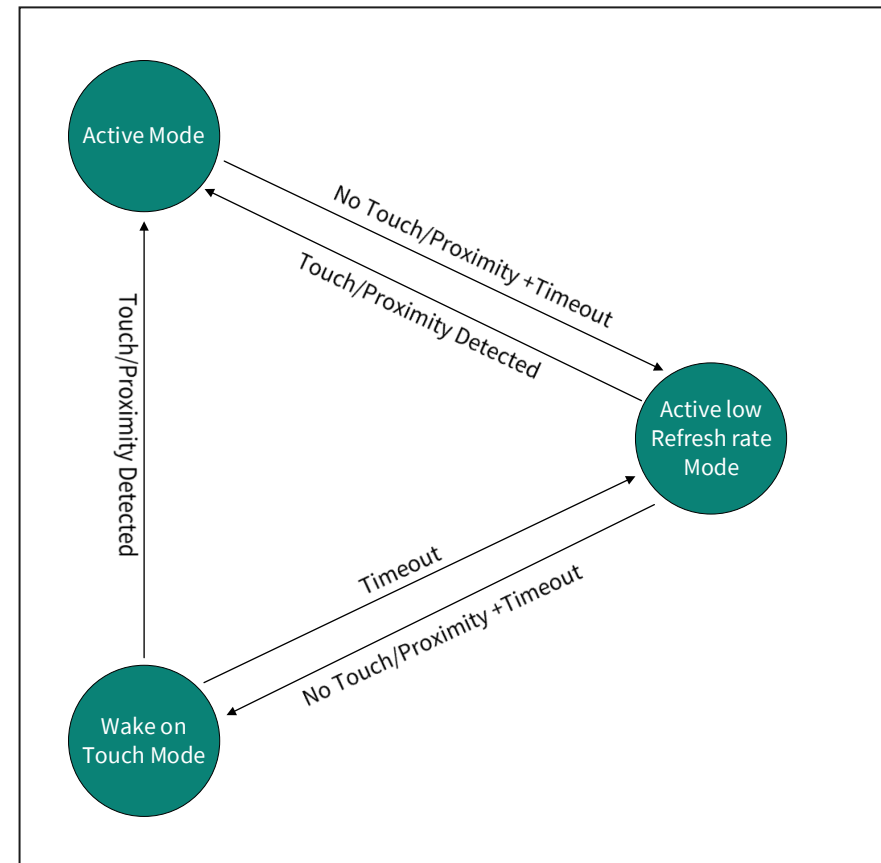


MSCLP – Low-power features

- MSCLP uses wake-on touch (WoT) mode to detect touch while the device is in Deep Sleep
- Autonomously scans several sensors without CPU intervention
 - The MSCLP has a dedicated 1024 bytes of SRAM
 - Active mode sensors
 - 21 sensors with HW IIR filter enabled
 - 36 sensors with HW IIR filter disabled
 - WoT mode sensors
 - 21 sensors HW IIR filter is always enabled for LP scanning
- Does not require the main IMO clock for scanning
 - Requires CAPSENSE™ IMO
 - Main IMO is required for gestures like click, scroll, flick, swipe, and rotate
- Reduces power in the Active mode
- Enables 5-10x lower-power consumption in the Low-power mode
- Enables 3-5x lower-power consumption in the Active mode

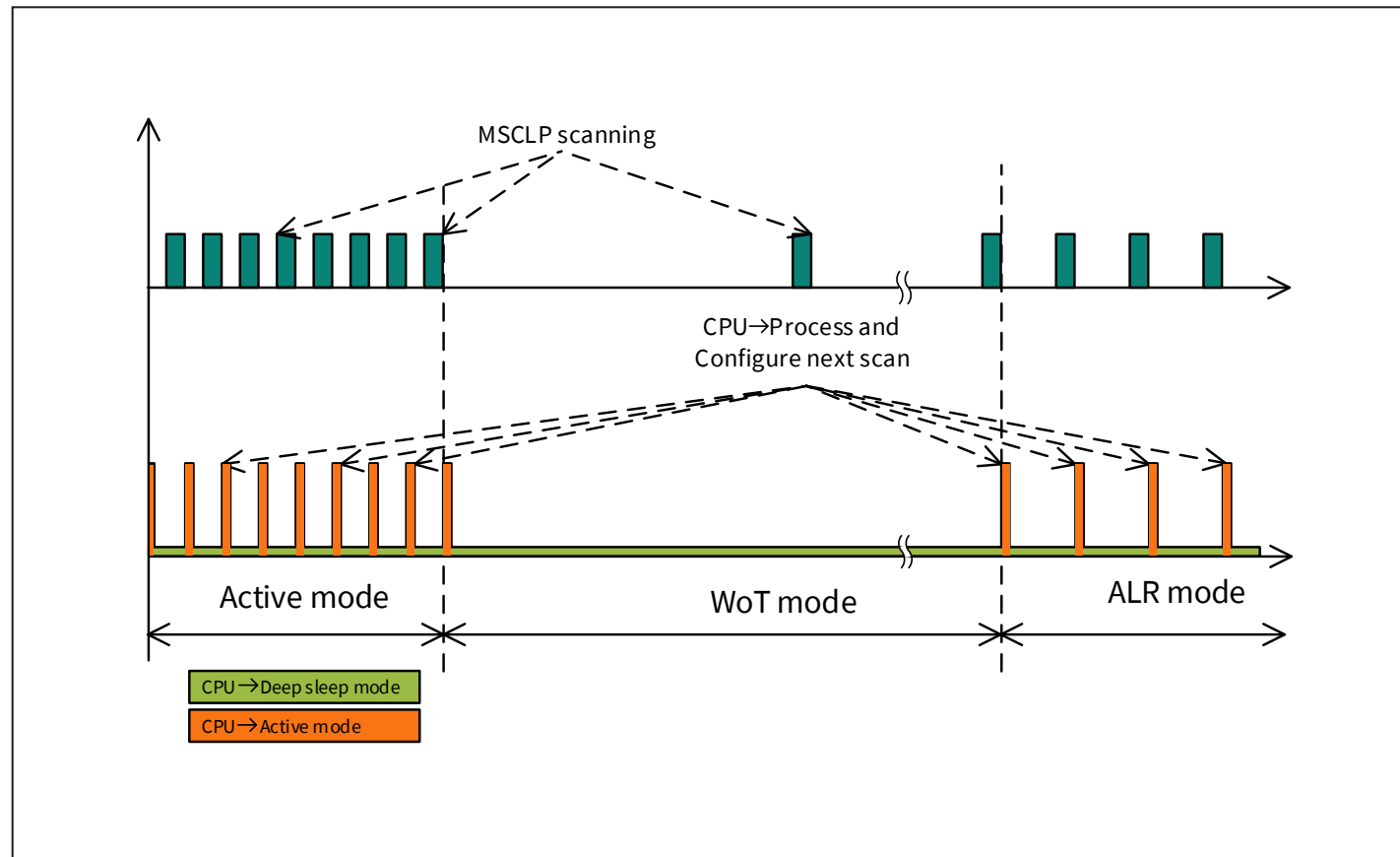
Application power modes

Application power modes	Scope	Refresh rate	Power consumption	Optimum device power mode
Active (max refresh rate)	All sensors	High	High	Active mode: To trigger scan and process Deep sleep mode: For frame scan and waiting (to achieve required refresh rate).
Active - low refresh rate (ALR)	All sensors	Low	Medium	Active mode: To trigger scan and process Deep Sleep mode: For frame scan and waiting (to achieve required refresh rate).
Wake on touch (WoT)	Few sensors or ganged sensors	Very Low	Low	Deep sleep mode: To scan, process, and wait.



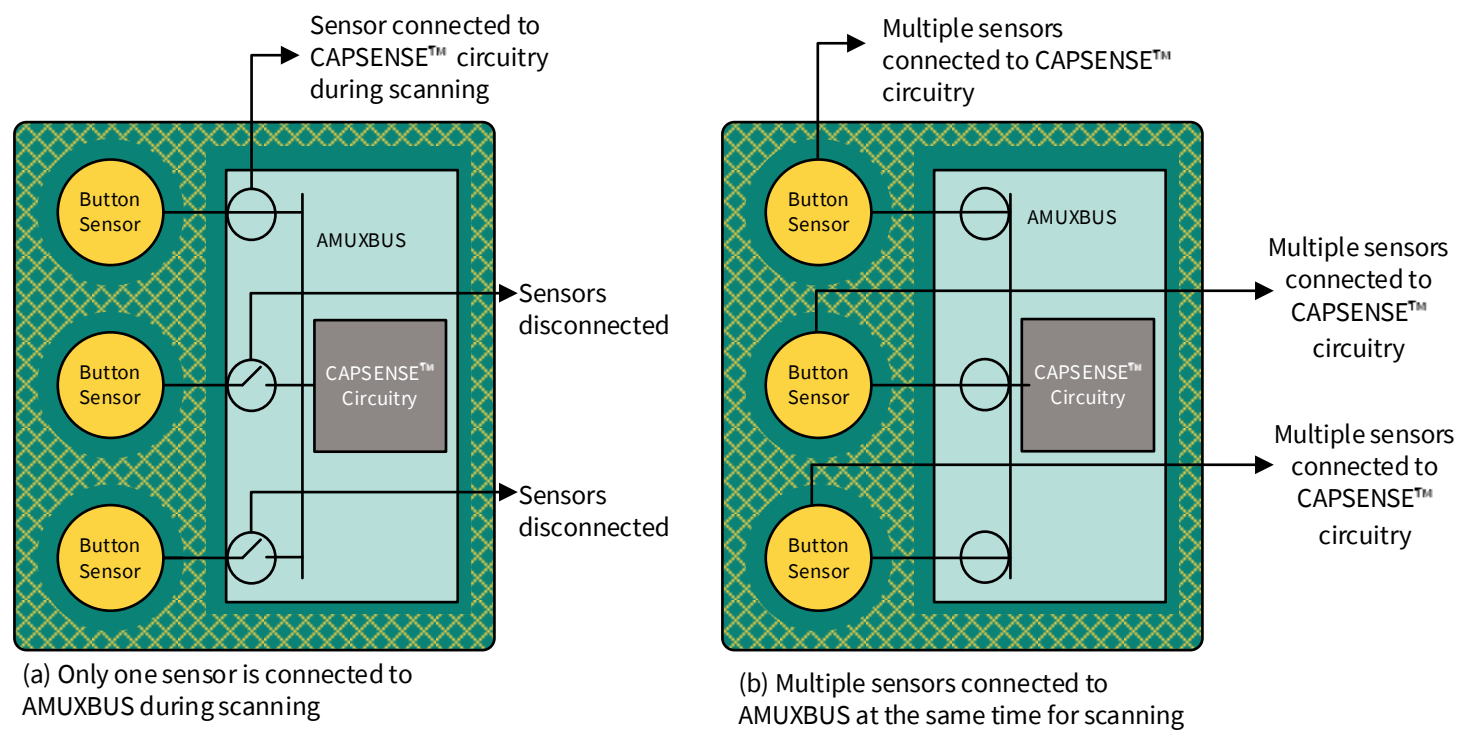
Wake-on-touch (WoT) sensing

- Autonomous scanning mode is the default scanning mode in fifth-generation LP CAPSENSE™
- Even in Active application power modes, the device power mode can be put to deep sleep for optimal power consumption
- Sensors can be ganged in the WoT mode to create a larger proximity sensor



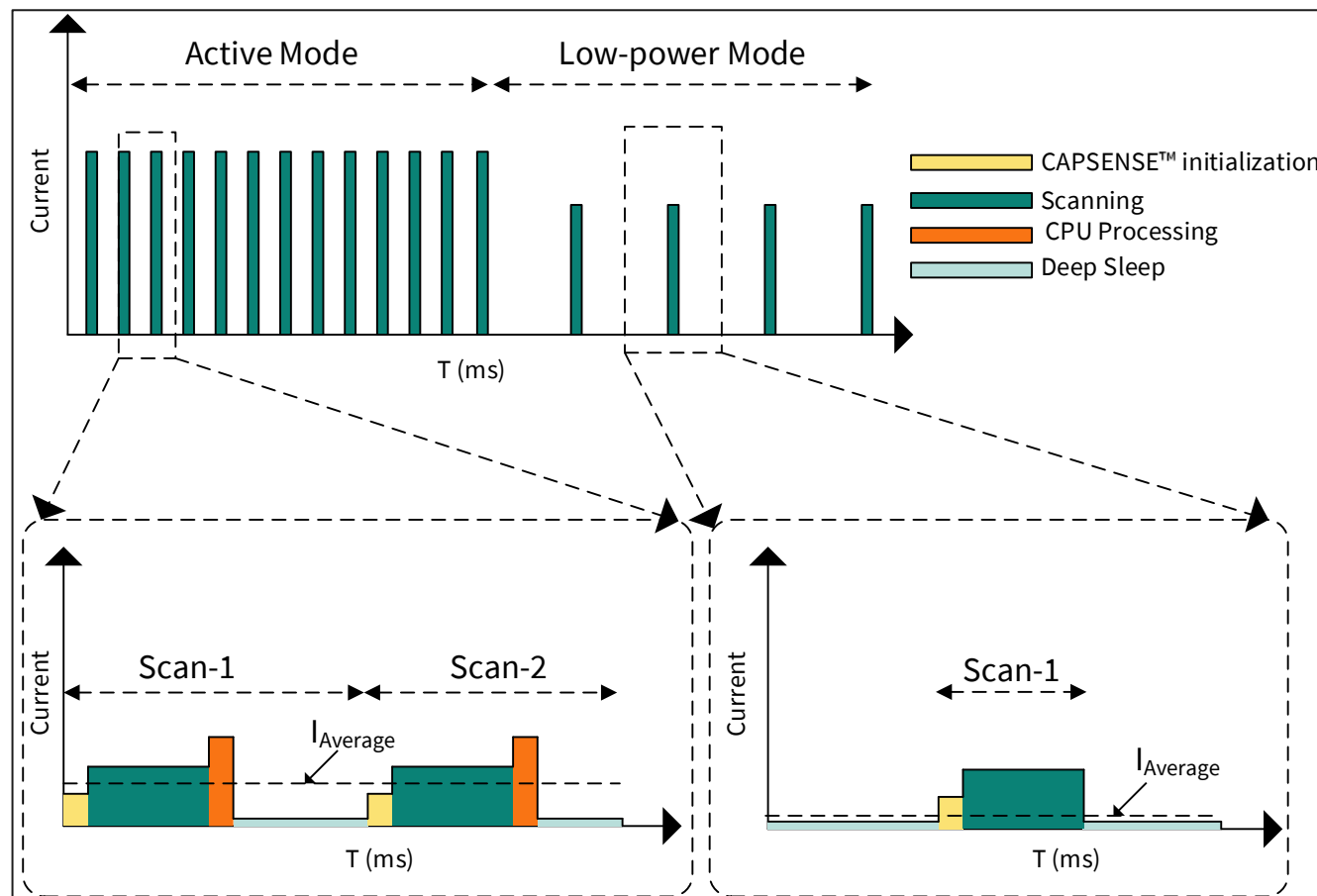
Sensor ganging

- Sensors can be ganged to create a larger proximity sensor
- This is especially beneficial for low-power sensors



Achieve low-power

- Scan duration
 - Depends on the C_p of the sensor
 - Max scan frequency limited by parasitic capacitance (C_p)
- Refresh rate and power modes
 - User experience and power
- Number of sensors in the low-power mode
 - Low C_p reduces power
 - Dedicated wakeup button is ideal
 - Gang sensors help to reduce power

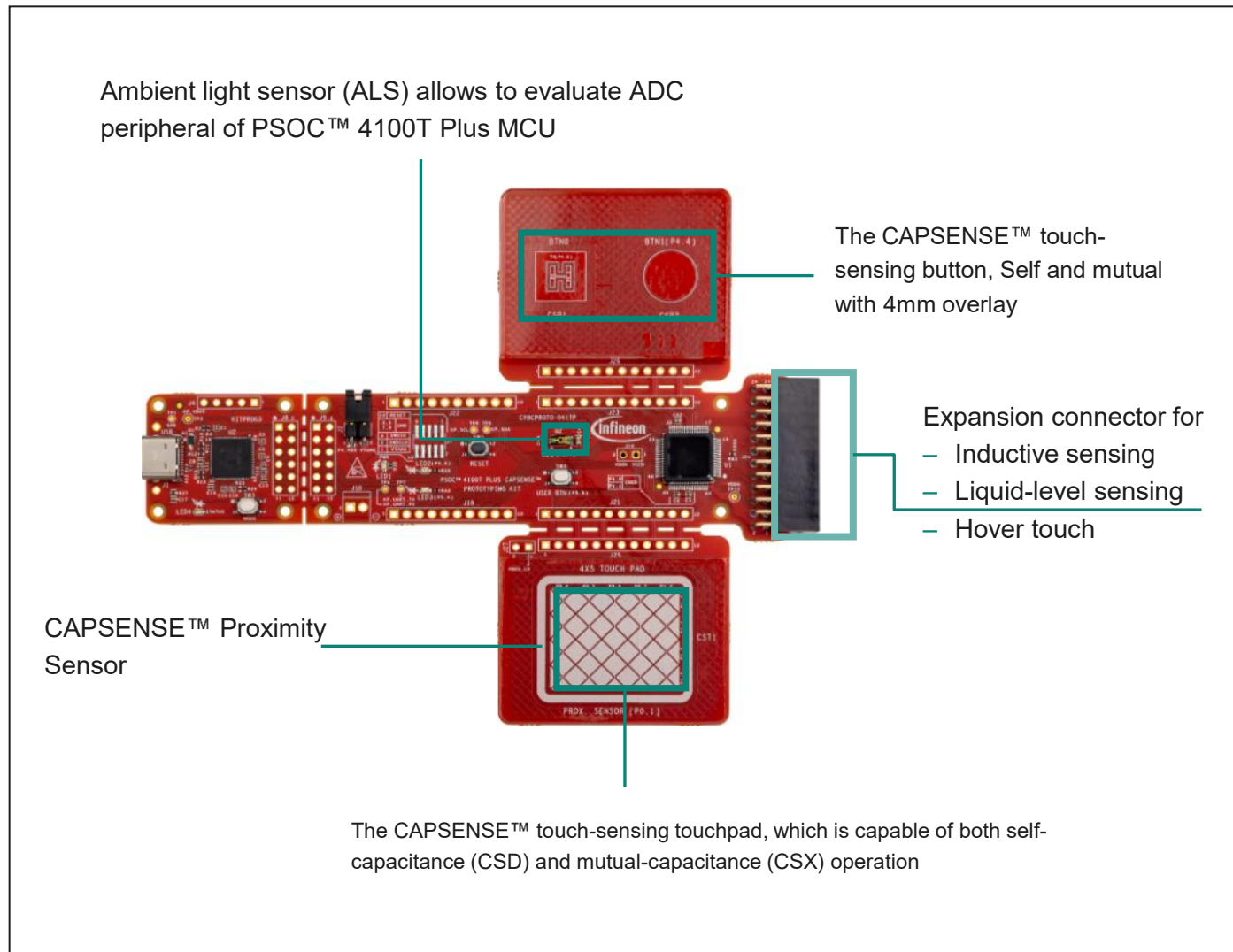


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CY8CPROTO-041TP - PSOC™ 4100T Plus Prototyping kit

Kit MPN	CY8CPROTO-041TP
Release schedule	Q2'2026 (Full Market Launch)
KitProg3	Yes
Autonomous operation without CPU	Yes
Ultra-low power Always-ON sensing	Yes
ADC measurements	Yes
CAPSENSE™ buttons	One self-cap button, one mutual-cap button
CAPSENSE™ trackpad	5 x 4 Touchpad, self and mutual sensing capable
CAPSENSE™ proximity	Yes
Inductive sensing	Through expansion board
Liquid-level sensing	Through expansion board
Expansion connectors	Yes, develop custom inductive and LLS boards or connect to expansion board of CY8CPROTO-040T-MS kit
Example code	CE240380, OOB example code
Best suited for	Traditional proto-typing kit that is easy to use, 32-bit MCU application development, connect external sensors on your own hardware and enables quick proof of concept development and evaluation of 5th Gen CAPSENSE™, Inductive Sensing and Liquid-level sensing



CY8CPROTO-041TP PSoC™ 4100T Plus CAPSENSE™ Prototyping Kit

– Low-power

- After reset, the application transitions to low-power states if there is no user activity to reduce the power consumption
- If there is touch detection while in low-power states, the application transitions to active mode with the highest refresh rate

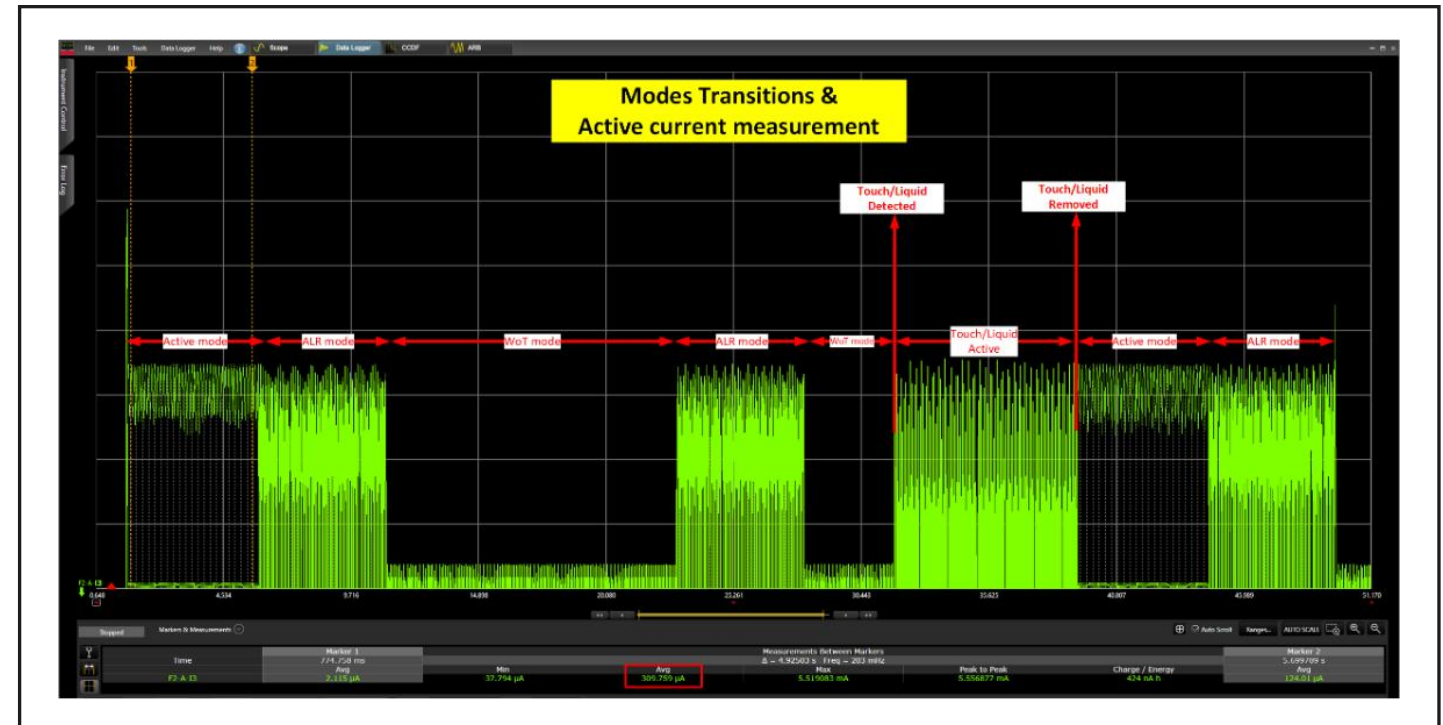
Device current at different application states

Application State	Refresh rate (Hz)	Current (μA)*
WoT	16	5
Active Low Refresh Rate	32	26
Active	128	96

* - For PSoC™ code example

Note: For more information on the specifics of the code example, see the [readme](#) of the code example.

Power consumption profile of PSoC™ 4100T Plus



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Lab 1: CAPSENSE™ performance tuning

What will you learn?

- How to set the initial hardware parameters
- How to set the sense clock frequency
- How to fine tune for the required SNR, power, and refresh rate
- How to tune the threshold parameters

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Lab 2: CAPSENSE™ low-power tuning

What will you learn?

- How to use Deep Sleep in regular widget scanning
- How to optimise the refresh rate and power consumption
- How to use application power states to conserve power

Revision history

Document revision	Date	Description of change
**	2026-12-04	Initial release.
*A	2026-05-19	Updated to the latest empower presentation template.



Thank you! Stay tuned for more trainings...

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